PAICS Deficiency — Comprehensive Disease Characterization Report

Category: Mendelian (autosomal recessive inborn error of de novo purine synthesis) Report compiled 2026-09-02. Evidence source types: human clinical case reports, in vitro/enzymatic studies, structural biology, and comparison with allied de novo purine synthesis (DNPS) disorders.

> **Caveat on evidence base:** PAICS deficiency is an *ultra-rare* Mendelian disorder. As of 2025 only **three molecularly confirmed patients** (plus one antenatally suspected sibling) have been reported worldwide, all homozygous for the same recurrent variant. Consequently, most disease-level statements rest on 1–3 individual patients (individual-patient/case-report evidence, not aggregated registry data). Where information is absent, this is stated explicitly. Many entries are extrapolated from the better-characterized allied DNPS defects (ADSL deficiency, AICA-ribosiduria/ATIC deficiency) and are flagged as such.

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1. Disease Information

Overview. PAICS deficiency is an autosomal recessive **inborn error of de novo purine synthesis (DNPS)** caused by biallelic loss-of-function variants in the *PAICS* gene. *PAICS* encodes a bifunctional enzyme — **phosphoribosylaminoimidazole carboxylase (AIR carboxylase, AIRC; EC 4.1.1.21)** and **phosphoribosylaminoimidazole-succinocarboxamide synthetase (SAICAR synthetase, SAICARS; EC 6.3.2.6)** — that catalyzes the sixth and seventh of the ten sequential steps that convert PRPP to inosine monophosphate (IMP). Clinically the disorder manifests as a **multiple congenital malformation (polymalformative) syndrome**. The two index cases died in the early neonatal period; the third reported patient survived with a milder, non-lethal course and normal neurodevelopment, indicating a broader phenotypic spectrum than originally recognized (P 31600779; P 39726239).

Key identifiers. - **Gene:** *PAICS* — HGNC:8587; NCBI Gene ID 10606; UniProt **P22234**; OMIM gene **172439**; Ensembl ENSG00000128050. Genomic locus **chr4:56,435,741–56,464,578 (GRCh38)**, cytoband **4q12** *(verified via gnomAD).* - **Disease (phenotype) — VERIFIED IDENTIFIERS: MONDO:0859003**; OMIM **#619859** ("Phosphoribosylaminoimidazole carboxylase deficiency"); **Orphanet ORPHA:633099**; **GARD:0026646** *(all confirmed via OLS4/ Monarch and HPO/JAX APIs)*. ICD-10: no specific code (closest E79.8 "Other specified disorders of purine and pyrimidine metabolism"); ICD-11: closest 5C55.2 (inborn errors of purine or pyrimidine metabolism). MeSH: no dedicated descriptor; indexed under "Purine-Pyrimidine Metabolism, Inborn Errors." - *Note: an earlier draft cited OMIM #618121; the correct phenotype MIM is #619859.* - **EC numbers:** 4.1.1.21 (AIRC) and 6.3.2.6 (SAICARS).

Synonyms / alternative names. - Phosphoribosylaminoimidazole carboxylase deficiency - SAICAR synthetase deficiency / AIR carboxylase deficiency - Multiple malformations syndrome, lethal, due to PAICS deficiency - Inborn error of de novo purine synthesis, PAICS type

Data provenance. Disease-level knowledge is derived from **individual patient case reports** (n=3), supporting enzymatic/cell-biology experiments in patient fibroblasts and recombinant protein, and CRISPR HeLa models — *not* from EHR-scale or registry aggregation.

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2. Etiology

Primary cause (genetic). Biallelic (homozygous, in a consanguineous/founder context) pathogenic variants in *PAICS*, producing a hypomorphic bifunctional enzyme. All reported patients carry the **homozygous missense variant NM_006452.4:c.158A>G, p.(Lys53Arg)**, which reduces catalytic activity (patient fibroblasts ~10% of control; recombinant enzyme ~25% of wild-type; carriers ~50%) (P 31600779).

Genetic risk factors. - *Causal variant:* PAICS c.158A>G p.(Lys53Arg) — recurrent across all reported families. - *Consanguinity / recurrent European allele:* the two index patients were from a consanguineous **Faroe Islands** family. The recurrent p.(Lys53Arg) allele (rs192831239) is a **low-frequency pan-European variant** (gnomAD NFE AF 0.135%), so its appearance in the unrelated French third case reflects the allele's general European frequency rather than a private founder. Consanguinity remains a strong risk factor for homozygosity, as for most recessive DNPS defects. - *Modifier genes:* none identified; the marked phenotypic difference (lethal vs. surviving with normal cognition) among patients homozygous for the *same* variant implies the existence of unknown genetic and/or environmental modifiers (P 39726239).

Environmental / lifestyle risk factors. None established. As a Mendelian congenital disorder, environmental exposure is not a primary driver. (Not applicable / no evidence.)

Protective factors. None documented. Heterozygous carriers (~50% residual activity) are asymptomatic, indicating ~50% enzyme activity is sufficient (haplosufficiency), consistent with recessive inheritance.

Gene–environment interactions. No data. Speculatively, dietary purine intake or salvage-pathway flux could modulate severity (as purine salvage can partly compensate for DNPS defects), but this is untested in PAICS deficiency.

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3. Phenotypes

Phenotype data derive from three patients. The two Faroese siblings (P 31600779) had a **lethal neonatal multiple-malformation** presentation; the third patient (P 39726239) had a **non-lethal polymalformative syndrome with normal neurodevelopment**, expanding the spectrum.

Official curated HPO annotations with frequencies (from OMIM #619859 / HPO-JAX; frequencies reflect the 2 index siblings, n=2, unless noted). These are the authoritative phenotype associations for knowledge-base ingestion:

HPO term	Phenotype	System	Frequency
HP:0001561	Polyhydramnios	prenatal	2/2
HP:0011461	Fetal onset	clinical course	2/2
HP:0003811	Neonatal death	clinical course	2/2
HP:0012368	Flat face	craniofacial	2/2
HP:0000248	Brachycephaly	craniofacial	2/2
HP:0003196	Short nose	craniofacial	2/2
HP:0005280	Depressed nasal bridge	craniofacial	2/2
HP:0000470	Short neck	head/neck	2/2
HP:0000369	Low-set ears	ear	2/2
HP:0002032	Esophageal atresia	digestive/ foregut	2/2
HP:0002575	Tracheoesophageal fistula	digestive/ foregut	1/2
HP:0000453 / HP:0000452 / HP:0004502	Choanal atresia / stenosis / bilateral choanal atresia	head/neck	1/2
HP:0000463	Anteverted nares	craniofacial	1/2
HP:0000316	Hypertelorism	eye	1/2
HP:0004322	Short stature	growth	1/2
HP:0001762	Talipes equinovarus	limbs	1/2
HP:0004209	Clinodactyly of the 5th finger	limbs	1/2
HP:0000921	Missing ribs	skeletal	1/2
HP:0008439	Lumbar hemivertebrae	skeletal	1/2
HP:0008743	Coronal hypospadias	genitourinary	1/1 (male)
HP:0008689	Bilateral cryptorchidism	genitourinary	1/1 (male)
HP:0000007	Autosomal recessive inheritance	inheritance	—

Additional features from the third (surviving) patient (39726239), not in the OMIM/HPO n=2 curation: congenital heart defect/cardiopathy (HP:0001627, newly described), plus **preserved/normal neurodevelopment** (distinguishing feature). The MONDO/OMIM narrative also lists nasal hypoplasia and lung malformations.

Phenotype characteristics. - *Age of onset:* congenital/neonatal (prenatal in the antenatally diagnosed sibling). - *Severity:* **variable** — lethal neonatal to survivable childhood form. - *Progression:* congenital and largely static (malformations fixed); the survivor showed normal neurodevelopmental trajectory. - *Frequency among affected:* with n=3, "frequencies" are indicative only — skeletal and oesophageal defects appear consistent (reported as a recurring theme); congenital heart disease in 1/3; early death in 2/3.

Quality-of-life impact. In the lethal form, QoL is dominated by neonatal demise. In the survivor, impact relates to surgical correction of malformations (cardiac, oesophageal, skeletal) with preserved cognition — a comparatively favourable functional outlook (39726239). No formal EQ-5D/SF-36/PROMIS data exist.

Note: Allied DNPS disorders (ADSL, ATIC) are dominated by **neurological** features (psychomotor retardation, epilepsy, autistic features, visual impairment) (25112391; 32557644). Notably, PAICS deficiency in the survivor spared the CNS, distinguishing it phenotypically.

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4. Genetic / Molecular Information

Causal gene. *PAICS* (HGNC:8587; OMIM 172439; Gene ID 10606; UniProt P22234), chromosome **4q12**. Encodes a **425-aa bifunctional polypeptide** that assembles into a homo-octamer (17224163).

Pathogenic variant(s). - **NM_006452.4:c.158A>G, p.(Lys53Arg)** (= NM_001079524.2:c.158A>G) — missense; homozygous in all reported patients. **ClinVar: Likely pathogenic** (review status "criteria provided, multiple submitters, no conflicts", verified via NCBI eutils). Affects the structure of the enzyme's catalytic site (P 31600779). - *PAICS* has **81 ClinVar entries**, overwhelmingly **VUS**; besides p.Lys53Arg, a truncating variant **c.843_844del (p.Cys281_Glu282delinsTer)** is also listed as Likely pathogenic (no published clinical report identified) — indicating additional candidate pathogenic alleles may exist beyond the recurrent founder missense. - Variant type/class: **missense** (single-nucleotide substitution). - Functional consequence: **loss of function (hypomorph)** — residual ~10–25% activity; reduces flux through both AIRC and SAICARS reactions; abolishes purinosome assembly. - **Allele frequency (gnomAD v4, verified):** dbSNP **rs192831239**; SPDI NC_000004.12:56441803:A:G (**GRCh38 chr4:56,441,804 A>G**); OMIM allelic variant 172439.0001; ClinGen CA2930671. Exome **AC 1675 / AN 1,448,934, AF 0.116%**, **0 homozygotes**; genome AF 0.072%, 0 homozygotes. Ancestry: highest in **non-Finnish European (0.135%)**, then South Asian (0.079%), Admixed American (0.070%), Finnish (0.025%), African (0.018%); absent in Ashkenazi Jewish, East Asian, Middle Eastern. → The allele is a **recurrent low-frequency pan-European variant, NOT a private Faroese founder mutation**; the complete absence of homozygotes (despite carrier frequency ~1/370–1/740 in Europeans) is consistent with recessive prenatal/neonatal lethality removing homozygotes from population databases (and possible under-ascertainment/reduced penetrance). - Somatic vs. germline: **germline** (congenital, biallelic).

gnomAD gene-level constraint (verified). *PAICS* is **not** loss-of-function-intolerant: **pLI ≈ 0** (5.8×10^{-7}), **LOEUF 0.82** (oe_lof 0.61; observed/expected LoF 33/53.7), missense z = 1.92 (mild constraint). This lack of haploinsufficiency is fully consistent with **autosomal recessive** inheritance — a single functional allele suffices (heterozygotes are asymptomatic with ~50% activity).

ACMG considerations. p.(Lys53Arg) is classified **Likely pathogenic in ClinVar** (multiple submitters, no conflicts). Supporting criteria: strong functional evidence (PS3 — measured reduced enzyme activity and abolished purinosome assembly, rescued by wild-type PAICS), rarity (PM2), and recurrence across unrelated families. Beyond this founder missense, most *PAICS* ClinVar entries are VUS, with one additional Likely-pathogenic truncating allele (c.843_844del).

Modifier genes. None identified; presence strongly inferred from intra-genotype phenotypic variability (lethal vs. surviving).

Epigenetic information. No disease-specific methylation/chromatin data for PAICS deficiency. (In cancer biology, *PAICS* is subject to m6A-mediated and H3K9me3/HP1α-linked regulation of the ASB11 axis controlling purinosome assembly —

P 37848033,

P 42493545

— but this is oncologic, not germline-disease, context.)

Chromosomal abnormalities. None; PAICS deficiency is a single-gene point-mutation disorder. (In cancers, chromosome-4q loss reduces PAICS expression —

P 33596246

— unrelated to the Mendelian disease.)

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5. Environmental Information

- **Environmental factors / toxins / radiation:** none implicated. (Not applicable.)
- **Lifestyle factors:** none implicated; congenital genetic disorder.
- **Infectious agents:** none; not an infectious/triggered disease.

The only "environmental" dimension of theoretical relevance is dietary purine availability and salvage-pathway substrate supply, which could in principle modulate a DNPS defect, but no evidence exists in PAICS deficiency.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Homozygous PAICS c.158A>G (p.Lys53Arg)** alters the catalytic-site structure of the bifunctional AIRC/SAICARS enzyme → **leads to** reduced catalytic activity (~10–25% residual) [demonstrated: enzyme assays,

   P 31600779].

2. Reduced PAICS activity **results in** a block at steps 6–7 of de novo purine synthesis (AIR → CAIR → SAICAR) → impaired conversion toward IMP [demonstrated biochemically in DNPS defects; P 35323684].
3. The mutant PAICS **fails to nucleate the purinosome** (the multi-enzyme DNPS metabolon); purinosome assembly is abolished and is rescued only by wild-type PAICS [demonstrated in patient fibroblasts, P 31600779 ; PAICS is a hub of purinosome protein–protein interactions, P 35331738].
4. Loss of channeled DNPS **leads to** two downstream, non-exclusive insults:
5. **(4a) Purine nucleotide insufficiency** → reduced supply of AMP/GMP/ATP/GTP for nucleic-acid synthesis and energy/signaling → impaired proliferation during rapid embryonic morphogenesis → **congenital malformations** [inferred].
6. **(4b) Accumulation of upstream intermediates** (predicted AIR/CAIR; their dephosphorylated ribosides, e.g., AIr) → **cytotoxicity** (AIr shown cytotoxic to multiple cell lines) → cell death/dysmorphogenesis [partly demonstrated: AIr toxicity, P 31600779 ; analogous to SAICAr/S-Ado toxicity in ADSL and AICA-riboside toxicity in ATIC deficiency, P 25112391 , P 32557644].
7. Tissue-level consequences during organogenesis **result in** structural defects of the **skeleton, oesophagus (foregut), and heart**, plus growth restriction and dysmorphism → the **polymalformative syndrome**; where the insult is severe, **early neonatal death** [observed, P 31600779 ; P 39726239].
8. **Branch point:** with only partial enzyme loss and/or protective modifiers, purine supply may be sufficient postnatally to permit **survival with normal neurodevelopment** (third

patient) — the CNS may be relatively spared compared with ADSL/ATIC defects [observed, P 39726239; mechanism of sparing inferred].

Molecular pathways. De novo purine biosynthesis (KEGG hsa00230 purine metabolism; Reactome "Purine ribonucleoside monophosphate biosynthesis"). PAICS catalyzes: AIR + CO_2 → CAIR (AIRC, EC 4.1.1.21) and CAIR + L-aspartate + ATP → SAICAR (SAICARS, EC 6.3.2.6).

Cellular processes. Metabolon (purinosome) assembly/phase separation; nucleotide biosynthesis; cell proliferation; apoptosis (from intermediate cytotoxicity). GO suggestions: GO:0006189 ('de novo' IMP biosynthetic process), GO:0009152 (purine ribonucleotide biosynthetic process), GO:0034023 (purinosome — as protein complex/assembly context), GO:0004638 (phosphoribosylaminoimidazole carboxylase activity), GO:0004639 (phosphoribosylaminoimidazolesuccinocarboxamide synthase activity).

Protein dysfunction. p.Lys53Arg is a loss-of-function/hypomorphic substitution distorting the catalytic site; the enzyme normally functions only as an **octamer** with substrate-channeling tunnels between AIRC and SAICARS active sites (P 17224163; P 32571877; reaction mechanism, P 35914774). Loss of activity + loss of purinosome-nucleating protein–protein interactions (P 35331738).

Experimental structures (RCSB PDB): 2H31 (octameric apo structure, P 17224163), 6YB8 / 6YB9 (substrate/product complexes, P 32571877), 7ALE; UniProt **P22234**; AlphaFold model **AF-P22234-F1**. Residue Lys53 lies in the AIR-carboxylase domain near the catalytic site.

Metabolic changes. Amino-acid/nucleotide metabolism: reduced IMP→AMP/GMP; predicted accumulation of AIR/CAIR/ SAICAR and their ribosides (CHEBI: aminoimidazole ribotide/AIR; SAICAR). Metabolomic profiling of DNPS-deficient HeLa cells shows accumulation of intermediates immediately upstream of the deficient enzyme (P 35323684).

Immune involvement. None described (not an immunologic disease).

Tissue-damage mechanisms. Cytotoxicity of accumulated dephosphorylated intermediates; nucleotide starvation of proliferating cells.

Molecular profiling. CRISPR-Cas9 PAICS-knockout/deficient HeLa cells provide targeted + untargeted metabolomic signatures of the DNPS block (P 35323684, P 35331738). No patient transcriptomic/proteomic/metabolomic datasets published beyond fibroblast enzymology.

Cell types & GO/CL suggestions. Fibroblasts used experimentally (CL:0000057). In vivo affected cell populations are rapidly proliferating embryonic progenitors of skeletal (CL:0000062 osteoblast; CL:0000138 chondrocyte), cardiac (CL:0000746 cardiac muscle cell), and foregut/oesophageal epithelium (CL:0000066 epithelial cell) — inferred from malformation pattern.

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7. Anatomical Structures Affected

Organ level (primary): skeleton (UBERON:0001434 skeletal system), oesophagus (UBERON:0001043), heart (UBERON:0000948). Growth (whole-body/UBERON multi-organ). **Secondary:** consequences of malformations (e.g., feeding/airway from oesophageal defects; circulatory from cardiac defects).

Body systems: musculoskeletal, digestive (foregut), cardiovascular; generalized growth. CNS **relatively spared** in the survivor (contrast with allied DNPS disorders that are CNS-dominant).

Tissue/cell level: connective/skeletal tissue (cartilage, bone), cardiac muscle, gut epithelium. CL terms as above.

Subcellular level: the purinosome is a **mitochondria-associated** cytoplasmic metabolon (P 35331738). GO cellular-component suggestions: GO:0005829 (cytosol), mitochondrial outer-membrane association; the purinosome itself is a dynamic, membraneless (phase-separated) body (P 37848033).

Localization / lateralization: malformations are congenital and can be midline/bilateral (skeletal, cardiac, oesophageal); no consistent lateralization reported (n too small).

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8. Temporal Development

- **Onset:** congenital; detectable **antenatally** (a suspected sibling recurrence was diagnosed prenatally, P 39726239). Onset pattern: **chronic/congenital** (present from organogenesis).
- **Progression / course:** malformations are structurally fixed. In the lethal form, course is **acute neonatal deterioration → death within days** (P 31600779). In the surviving form, course is **stable** post-surgical correction with normal developmental trajectory to at least 7 years (P 39726239).
- **Disease duration:** lethal (self-limited by neonatal death) vs. chronic/lifelong in survivors.
- **Critical periods:** first-trimester organogenesis is the window of vulnerability (skeletal/foregut/cardiac morphogenesis). Prenatal detection offers a window for counseling; no fetal/neonatal metabolic intervention is established.

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9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic *PAICS* variants; heterozygous parents asymptomatic).
- **Epidemiology:** ultra-rare; **prevalence/incidence unknown** (<1/1,000,000; only 3 reported cases). No registry data.
- **Penetrance:** appears complete for the malformation phenotype in homozygotes, but **expressivity is highly variable** (lethal neonatal vs. surviving with normal cognition) despite the identical genotype (P 39726239).
- **Expressivity:** variable (see above).
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** not reported.
- **Founder effect / consanguinity (revised via gnomAD):** the recurrent p.(Lys53Arg)/ rs192831239 allele is **not a private Faroese founder** but a **recurrent low-frequency pan-European variant** (gnomAD exome AF 0.116%; highest in non-Finnish Europeans, 0.135%). Its recurrence in an unrelated French patient is therefore expected. The index family was consanguineous (Faroe Islands), which brought two copies together; consanguinity remains a risk factor for homozygosity of this and other recessive alleles.
- **Carrier frequency (verified): ~1/370–1/740 in Europeans** (2×AF ≈ 0.23–0.27% NFE); rarer in African, absent in Ashkenazi/East Asian/Middle Eastern gnomAD samples. Notably **zero homozygotes** are observed in ~1.45 million gnomAD alleles, despite an expected homozygote birth frequency on the order of ~1/550,000 (NFE) — consistent with recessive prenatal/neonatal lethality (affected individuals excluded from gnomAD) and/or reduced penetrance; implies the disorder is likely **under-ascertained** (unrecognized fetal losses/ neonatal deaths).
- **Population demographics / geography:** first cases Faroe Islands; third case reported from a European (French) centre — consistent with the allele's European distribution. No sex predilection evident (both sexes affected; third patient male). Age distribution: prenatal-to-childhood onset.

10. Diagnostics

Diagnostic approach. Because DNPS-intermediate biomarkers may be **undetectable** (AIR/AIr were not found in patient fibroblasts, P 31600779), diagnosis is primarily **molecular/genomic**.

- **Genetic testing (primary): whole-genome sequencing (WGS)** established the diagnosis in the third patient; **whole-exome sequencing (WES)** is equally appropriate; single-gene *PAICS* testing/targeted analysis of c.158A>G is useful where the founder allele is suspected. Purine-metabolism/inborn-error gene panels including *PAICS* are appropriate (GTR).
- **Biochemical/laboratory:** urine/plasma/CSF purine metabolite profiling (HPLC-MS) as used for ADSL (SAICAr, S-Ado) and ATIC (AICA-riboside) — but note **classic accumulating markers may be absent** in PAICS deficiency, limiting biochemical screening (P 31600779). Enzyme activity assay in cultured skin fibroblasts (reduced to ~10%) is confirmatory (LOINC/enzyme assay).
- **Functional/cellular:** purinosome-assembly assay in fibroblasts (absent, rescued by WT PAICS) — a research-grade functional confirmation.
- **Imaging:** prenatal/postnatal imaging (fetal ultrasound, echocardiography, skeletal survey, contrast oesophagram) to delineate malformations. RadLex terms as appropriate.
- **Histopathology:** no pathognomonic biopsy finding described.

Clinical criteria / differential diagnosis. No formal criteria. Differential includes other **DNPS defects** (ADSL deficiency, AICA-ribosiduria/ATIC, ADSS/ADSS1/2, ATIC, PRPS abnormalities) — distinguished by their **CNS-dominant** presentation and specific accumulating metabolites; also other polymalformative/VACTERL-spectrum syndromes (given vertebral, cardiac, oesophageal involvement) and chromosomal disorders (excluded by CMA/karyotype/sequencing).

Screening. Carrier/cascade testing for the familial p.(Lys53Arg) variant; prenatal molecular testing feasible (used in the suspected sibling). Not part of routine newborn screening.

11. Outcome / Prognosis

- **Survival/mortality:** in the index family, **2/2 siblings died in the early neonatal period** (P 31600779). The third patient **survived to at least 7 years** with normal neurodevelopment (P 39726239). Thus prognosis is **bimodal/variable**, ranging from neonatal-lethal to survivable childhood disease.
- **Morbidity/function:** in survivors, morbidity is driven by structural malformations requiring surgical correction (cardiac, oesophageal, skeletal); **cognition may be preserved**, an important prognostic distinction from ADSL/ATIC deficiencies.
- **Complications:** those of congenital heart disease, oesophageal malformation (feeding/respiratory), and skeletal anomalies.
- **Prognostic factors:** severity/number of malformations and (inferred) residual enzyme function/modifiers determine lethality. The 2025 report explicitly **updated prognosis to include the possibility of survival with normal neurodevelopment** (P 39726239).
- **QoL measures:** none formally applied.

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12. Treatment

No disease-specific/curative therapy exists. Management is **supportive and symptomatic** (NCIT: Supportive Care Therapy; NCIT:C15277 Supportive Care).

- **Pharmacotherapy:** none targeted; there is no established purine-replacement therapy for PAICS deficiency. (Theoretical strategies — dietary/purine salvage support, avoidance of intermediate accumulation — are unproven.)
- **Surgical/interventional:** correction of congenital malformations (cardiac surgery, oesophageal repair, orthopedic management) in survivors (NCIT clinical-intervention terms as applicable, e.g., Cardiac Surgery, Esophageal repair).
- **Supportive/rehabilitative:** neonatal intensive care; nutritional support; multidisciplinary follow-up.

- **Advanced/experimental therapeutics:** none in trials for PAICS deficiency. Gene therapy, enzyme replacement, or substrate-modulation approaches are conceptual only. *(Note: PAICS is being pursued as an oncology drug target — inhibitors to reduce purine synthesis in cancer —*

  P *37673296* ,

  P *34344987*
  — the opposite therapeutic direction from this deficiency.)
- **Pharmacogenomics:** not applicable.
- **Treatment outcomes / adverse events:** determined by surgical/critical-care outcomes; no drug outcome data.

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13. Prevention

- **Primary prevention: genetic counseling** for consanguineous families and known carriers; **carrier/cascade screening** for the familial p.(Lys53Arg) allele; reproductive options including **preimplantation genetic testing (PGT)** and **prenatal molecular diagnosis** (feasible, as demonstrated antenatally in the suspected sibling,

  P 39726239).
- **Secondary prevention:** prenatal imaging + molecular testing enables early detection and delivery planning; early surgical correction of malformations in survivors.
- **Tertiary prevention:** management of complications of congenital malformations.
- **Immunization / public-health / environmental interventions:** not applicable (Mendelian disorder).
- **Counseling:** recurrence risk 25% per pregnancy for carrier couples (autosomal recessive); NSGC/ACMG genetic-counseling frameworks apply.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs (verified via Alliance of Genome Resources, stringent):** *PAICS* is deeply conserved across metazoa and fungi. One-to-one/many orthologs:
- Mouse *Paics* — MGI:1914304; NCBI Gene 67054; NCBITaxon:10090
- Rat *Paics* — RGD:620066; NCBITaxon:10116
- Zebrafish *paics* — ZFIN:ZDB-GENE-030131-9762; NCBITaxon:7955
- *Drosophila melanogaster Paics* — FB:FBgn0020513; NCBITaxon:7227
- *Caenorhabditis elegans paic-1* — WB:WBGene00015116; NCBITaxon:6239
- *Saccharomyces cerevisiae ADE1* — SGD:S000000070; NCBITaxon:4932 (note: yeast splits the bifunctional activity — ADE1 = SAICAR synthetase, ADE2 = AIR carboxylase)
- *Xenopus tropicalis/laevis paics* — Xenbase
- Bacterial functional homologs: PurC/PurE/PurK. DNPS is among the most conserved biosynthetic pathways.
- **Natural disease in other species:** no naturally occurring PAICS-deficiency disease is catalogued in OMIA for companion animals/livestock (none found — *verify*).
- **Comparative biology:** the octameric bifunctional vertebrate PAICS contrasts with separate monofunctional bacterial/yeast enzymes; substrate channeling and purinosome assembly are conserved features studied to understand human disease.
- **Zoonotic/cross-species transmission:** not applicable (genetic, non-transmissible).

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15. Model Organisms

- **Cellular / in vitro models (principal): CRISPR-Cas9 PAICS-knockout / deficient HeLa cells** used for targeted + untargeted metabolomic profiling of the DNPS block (P 35323684) and for dissecting purinosome protein–protein interactions and metabolic channeling (crPAICS cells, P 35331738). **Patient skin fibroblasts** (primary) recapitulate reduced enzyme activity and absent

purinosome assembly, rescued by wild-type PAICS transfection
(P 31600779).

- **Recombinant enzyme:** E. coli-expressed wild-type and p.Lys53Arg PAICS for kinetic characterization
(P 31600779);
human PAICS crystal structures for mechanistic/structural study
(P 17224163 ,
P 32571877 ,
P 35914774).
- **Genetic model types available:** knockout/knockdown cell lines (CRISPR, shRNA — the latter widely used in cancer studies, e.g., AML,
P 34344987).
- **Mouse *Paics* knockout (IMPC, verified via IMPC solr):** homozygous null mice show **"preweaning lethality, complete penetrance"** and **"prenatal lethality prior to heart atrial septation"** (homozygote, complete penetrance) — i.e., *Paics* is an **essential gene** and complete loss is embryonically/prenatally lethal. No viable homozygous-null adults. Suggested MP terms: MP:0011100 (preweaning lethality, complete penetrance), prenatal-lethality MP terms. This is mechanistically important: it explains why **human patients carry a hypomorphic missense** (p.Lys53Arg, ~10–25% residual activity) rather than biallelic nulls, and the "prenatal lethality prior to heart atrial septation" parallels the **congenital cardiac defect** in the surviving human patient
(P 39726239).
- **Phenotype recapitulation:** cell models reproduce the biochemical/purinosome defect well; the mouse null recapitulates lethality/essentiality but (being a complete null) does not model the *hypomorphic* human malformation syndrome. **Limitation:** no published hypomorphic/knock-in *Paics* animal model (e.g., p.Lys53Arg knock-in) recapitulating the human polymalformative phenotype exists (*gap / future direction*).
- **Resources:** Cellosaurus (HeLa derivatives), MGI/IMPC (mouse *Paics*), ZFIN (zebrafish *paics*) for reagents.

Key Ontology Term Suggestions (summary)

- **Gene/protein:** *PAICS* (HGNC:8587, UniProt P22234); GO:0004638, GO:0004639, GO:0006189, GO:0009152.
- **Phenotypes (HPO):** HP:0002011/HP:0001263 (malformation), HP:0003811 (neonatal death), HP:0002032 (esophageal atresia), HP:0001627 (abnormal heart morphology), HP:0000924 (skeletal abnormality), HP:0001999 (facial dysmorphism), HP:0001511 (IUGR).
- **Cell types (CL):** CL:0000057 (fibroblast), CL:0000062 (osteoblast), CL:0000138 (chondrocyte), CL:0000746 (cardiomyocyte), CL:0000066 (epithelial cell).
- **Anatomy (UBERON):** UBERON:0001434 (skeletal system), UBERON:0001043 (esophagus), UBERON:0000948 (heart).
- **Chemicals (CHEBI):** AIR/aminoimidazole ribotide, CAIR, SAICAR, IMP; substrate CO_2, L-aspartate, ATP.
- **Disease (MONDO/OMIM): MONDO:0859003**; OMIM #**619859**; **Orphanet ORPHA:633099**; **GARD:0026646** (all verified).
- **Treatment (NCIT):** supportive care; surgical repair of congenital anomalies.

Supported vs. Refuted Statements

Supported (evidence-based): - PAICS deficiency is AR, caused by biallelic *PAICS* p.(Lys53Arg); loss-of-function reduces enzyme activity and abolishes purinosome assembly (P 31600779). *(Strong: enzymatic + cellular rescue.)* - Phenotype = congenital multiple-malformation syndrome; spectrum spans neonatal-lethal to survivable-with-normal-cognition (P 31600779, P 39726239). *(Moderate: n=3.)* - Accumulating intermediate ribosides (AIr) are cytotoxic; analogous to other DNPS defects (P 31600779;

;

P 25112391

).

P 32557644

(*Moderate.*)

Refuted / not supported: - That classic urinary DNPS metabolite markers reliably diagnose PAICS deficiency — **refuted**: predicted markers were undetectable in patient fibroblasts; molecular testing is required (

P 31600779

). - That PAICS deficiency is uniformly CNS-degenerative like ADSL/ATIC — **not supported**: the survivor had normal neurodevelopment (

P 39726239

).

Limitations & Future Directions

- Evidence rests on **3 patients** with the **same variant**; genotype–phenotype breadth, penetrance, carrier frequency, and epidemiology are essentially unknown.
- No validated whole-animal disease model; mechanism of malformation (nucleotide starvation vs. intermediate toxicity) not fully resolved for organogenesis.
- Priorities: identify additional patients/variants; establish biomarker(s); build animal models; test whether purine-salvage support alters outcome; clarify modifiers explaining lethal-vs-survivable divergence.

Prepared for disease knowledge-base population. Verified against primary databases: disease identifiers (MONDO:0859003, OMIM #619859, ORPHA:633099, GARD:0026646) via OLS4/Monarch; HPO annotations via HPO-JAX; variant classification (ClinVar VCV001686821, Likely pathogenic) and coordinates (rs192831239, chr4:56,441,804 GRCh38) via NCBI eutils; allele frequency (gnomAD v4 exome AF 0.116%, 0 homozygotes) and gene constraint via gnomAD API; orthologs via Alliance of Genome Resources; mouse-knockout lethality via IMPC; PDB structures via RCSB. Remaining items to confirm before ingestion: exact OMIA status (no natural animal disease found), and any newer case reports post-2025.

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